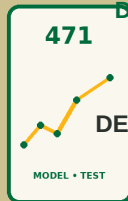
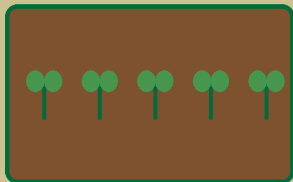


Define State and Control Variables



DEFINE STATE AND CONTROL VARIABLES

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

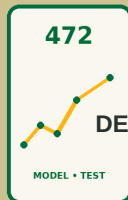
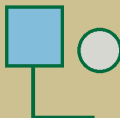
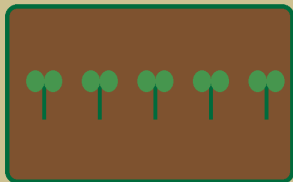
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Design Threshold Control



DESIGN THRESHOLD CONTROL

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

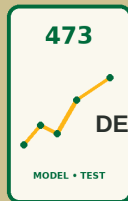
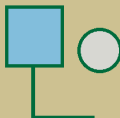
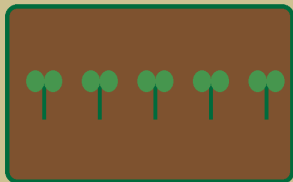
Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Design Hysteresis Control



DESIGN HYSTERESIS CONTROL

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

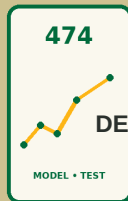
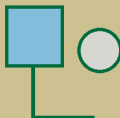
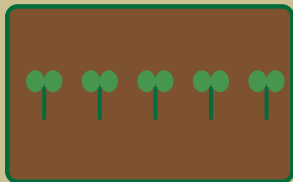
Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Map a Feedback Loop



MAP A FEEDBACK LOOP

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

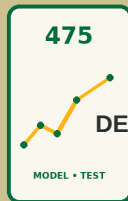
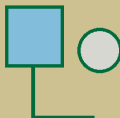
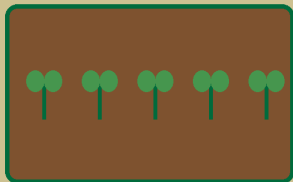
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Add a Feedforward Input



ADD A FEEDFORWARD INPUT

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

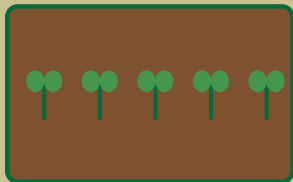
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EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Verify Irrigation Flow



VERIFY IRRIGATION FLOW

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

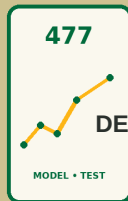
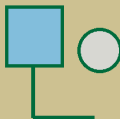
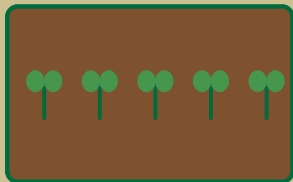
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Apply a Tank-Level Constraint



APPLY A TANK-LEVEL CONSTRAINT

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

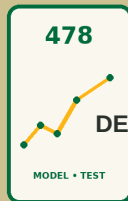
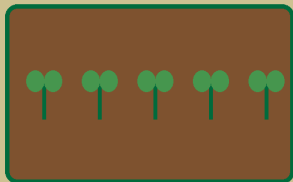
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Diagnose a Control Fault



DIAGNOSE A CONTROL FAULT

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

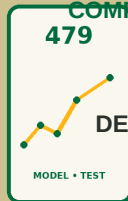
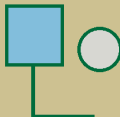
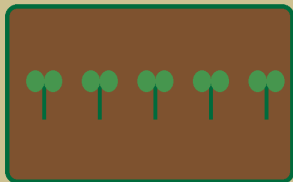
Identify the abnormal signal or performance pattern, formulate plausible causes, test them against independent evidence, and recommend a corrective action.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Compare Benchmark and Adaptive Control



COMPARE BENCHMARK AND ADAPTIVE CONTROL

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

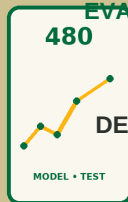
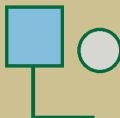
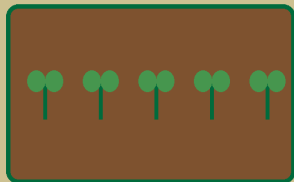
Use quantitative evidence and stated criteria. Compare alternatives or conditions, account for uncertainty, and justify the conclusion.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Evaluate Smart-Irrigation Performance



EVALUATE SMART-IRRIGATION PERFORMANCE

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

Use quantitative evidence and stated criteria. Compare alternatives or conditions, account for uncertainty, and justify the conclusion.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE